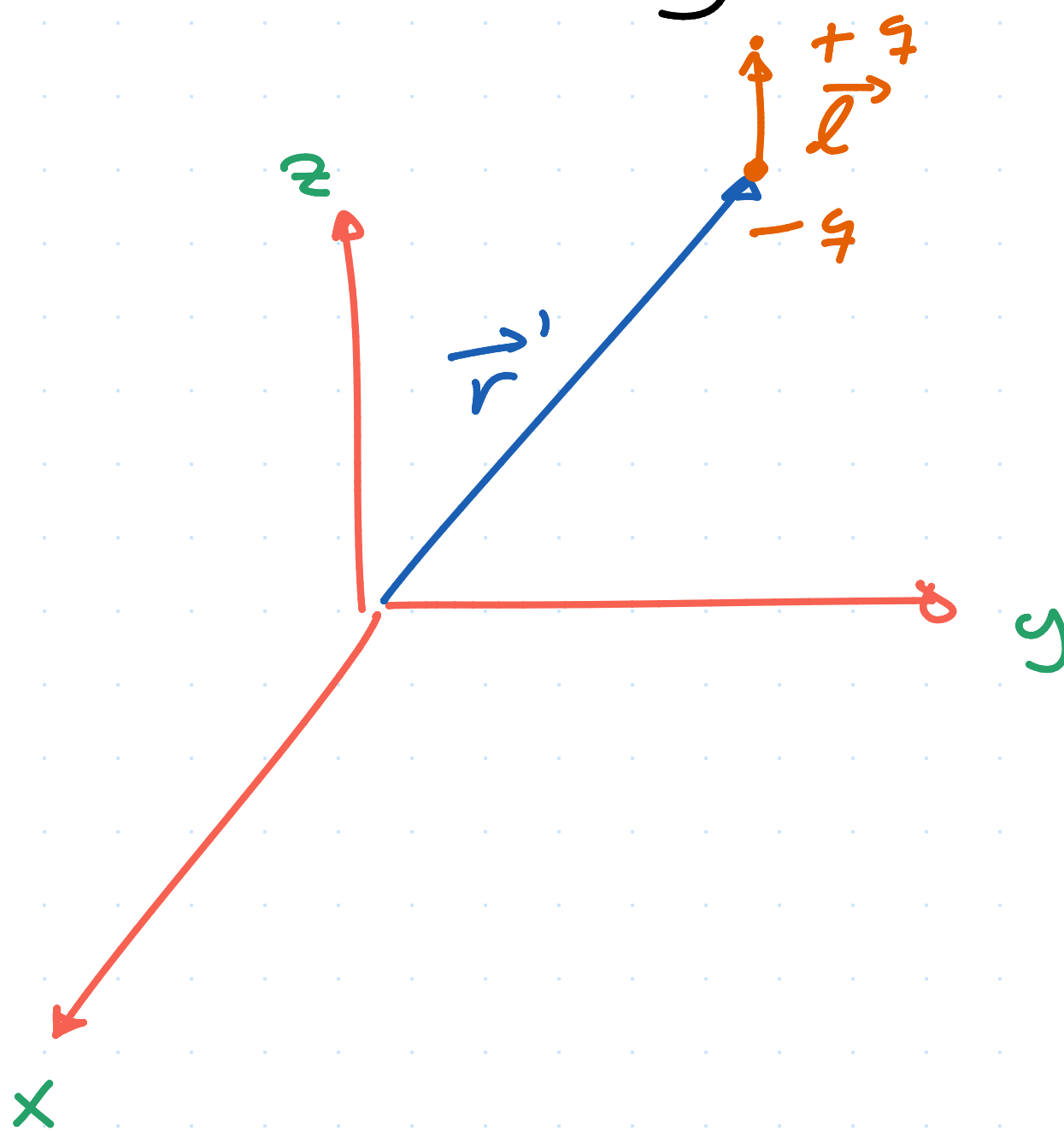


Unit 1: Electrostatics

$$(2) \quad \vec{E} = \frac{q}{4\pi\epsilon_0} \left[\frac{(\vec{r} - \vec{r}') - \vec{\ell}}{|\vec{r} - \vec{r}'|^3} \left[1 + \frac{3(\vec{r} - \vec{r}') \cdot \vec{\ell}}{|\vec{r} - \vec{r}'|^2} \dots \right] - \frac{\vec{r} - \vec{r}'}{|\vec{r} - \vec{r}'|^3} \right]$$

$$1) \quad \vec{E} = \frac{q}{4\pi\epsilon_0} \left[\frac{\vec{r} - \vec{r}' - \vec{\ell}}{|\vec{r} - \vec{r}' - \vec{\ell}|^3} - \frac{\vec{r} - \vec{r}'}{|\vec{r} - \vec{r}'|^3} \right]$$



$$\vec{E} = \frac{q}{4\pi\epsilon_0} \left[\frac{3(\vec{r} - \vec{r}') \cdot \vec{\ell} (\vec{r} - \vec{r}')}{|\vec{r} - \vec{r}'|^5} - \frac{\vec{\ell}}{|\vec{r} - \vec{r}'|^3} \right]$$

As $|\vec{\ell}| \rightarrow 0$, $q \rightarrow \infty$ such that $q\vec{\ell}$ is constant we

define the electric dipole moment as:

$$\vec{p} = q\vec{\ell}$$

$$\vec{E} = \frac{1}{4\pi\epsilon_0} \left[\frac{3(\vec{r} - \vec{r}') \cdot \vec{p} (\vec{r} - \vec{r}')}{|\vec{r} - \vec{r}'|^5} - \frac{\vec{p}}{|\vec{r} - \vec{r}'|^3} \right]$$

We know that

$$E = -\nabla u$$

$$V(\vec{r}) = \frac{q}{4\pi\epsilon_0} \left[\frac{1}{|\vec{r} - \vec{r}' - \vec{\ell}|} - \frac{1}{|\vec{r} - \vec{r}'|} \right]$$

Expandir:

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{2!} x^2$$

$$\frac{1}{|\vec{r} - \vec{r}' - \vec{\ell}|} =$$

Escalar a la forma $(1+x)^n$, cuando

$$f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \frac{f'''(a)}{3!}(x-a)^3 + \dots + \frac{f^{(n)}(a)}{n!}(x-a)^n$$

$$|\vec{r} - \vec{r}' - \vec{\ell}|^{-1} = \left[|\vec{r} - \vec{r}'|^2 - 2\vec{\ell} \cdot (\vec{r} - \vec{r}') + \vec{\ell} \cdot \vec{\ell} \right]^{-1/2}$$

Primero calculamos el producto punto:

$$|\vec{r} - \vec{r}' - \vec{\ell}|^2 = |\vec{r} - \vec{r}'|^2 - 2\vec{\ell} \cdot (\vec{r} - \vec{r}') + \vec{\ell} \cdot \vec{\ell}$$

$$|\vec{r}|^2 = \vec{r} \cdot \vec{r}; \quad |\vec{a} + \vec{b}|^2 = (\vec{a} + \vec{b})(\vec{a} + \vec{b}) = \vec{a} \cdot \vec{a} + 2\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{b}$$

$$|\vec{r} - \vec{r}' - \vec{\ell}|^{-1} = |\vec{r} - \vec{r}'|^{-1} \left\{ 1 - \frac{2\vec{\ell} \cdot (\vec{r} - \vec{r}')}{|\vec{r} - \vec{r}'|^2} + \frac{\vec{\ell} \cdot \vec{\ell}}{|\vec{r} - \vec{r}'|^2} \right\}^{-1/2}$$

$\alpha = -\frac{1}{2}$

$$f(x) = (1+x)^\alpha; \quad f'(x) = \alpha(1+x)^{\alpha-1};$$

$$f''(x) = \alpha(\alpha-1)(1+x)^{\alpha-2}$$

Tenemos

$$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \dots$$

$$(1+x)^\alpha = 1 + \alpha x + \frac{\alpha(\alpha-1)}{2!} x^2 + \dots +$$

$$F^{(1)}(\omega) \propto -\frac{1}{2} \left(1 - \frac{2 \vec{\ell} \cdot (\vec{r} - \vec{r}')}{|\vec{r} - \vec{r}'|^2} + \frac{\vec{\ell} \cdot \vec{\ell}}{|\vec{r} - \vec{r}'|^2} \right)^{-\frac{3}{2}} + -\frac{1}{2} \left(-\frac{3}{2} \right) \left(1 - \frac{2 \vec{\ell} \cdot (\vec{r} - \vec{r}')}{|\vec{r} - \vec{r}'|^2} + \frac{\vec{\ell} \cdot \vec{\ell}}{|\vec{r} - \vec{r}'|^2} \right)^{-\frac{5}{2}}$$

no se
toma en
cuenta

Tcemo

solo primer orden

$$|\vec{r} - \vec{r}' - \vec{\ell}|^{-1} = |\vec{r} - \vec{r}'|^{-1} \left\{ 1 + \frac{\vec{\ell} \cdot (\vec{r} - \vec{r}')}{|\vec{r} - \vec{r}'|^2} + \dots + \right\}$$

$$U(\vec{r}) = \frac{q}{4\pi\epsilon_0} \left[|\vec{r} - \vec{r}'| \left\{ 1 + \frac{\vec{\ell} \cdot (\vec{r} - \vec{r}')}{|\vec{r} - \vec{r}'|^2} + \dots + \right\} - \frac{1}{|\vec{r} - \vec{r}'|} \right]$$

Tcemo

$$U(\vec{r}) = \frac{1}{4\pi\epsilon_0} \frac{\vec{p} \cdot (\vec{r} - \vec{r}')}{|\vec{r} - \vec{r}'|^3}$$

Tarea subir una breve descripción de la sección
 ↑
 Fin de semana.

(2-39) a pág (2-44) ⁴¹⁵ ¹¹
 Dipole

An electric dipole

Taglo - en varios var

$$f(\vec{x} + \vec{h}) = f(\vec{x}) + \vec{h} \cdot \nabla f(\vec{x}) + \frac{\vec{h}^T \nabla^2 f(\vec{x}) \vec{h}}{2!} + \dots +$$

Desarrollo multipolar

Caso eléctrico a primer orden

momento dipolar de la
distribución:

↳ Desarrollo dipolar

↳ Desarrollo cuadrupolar

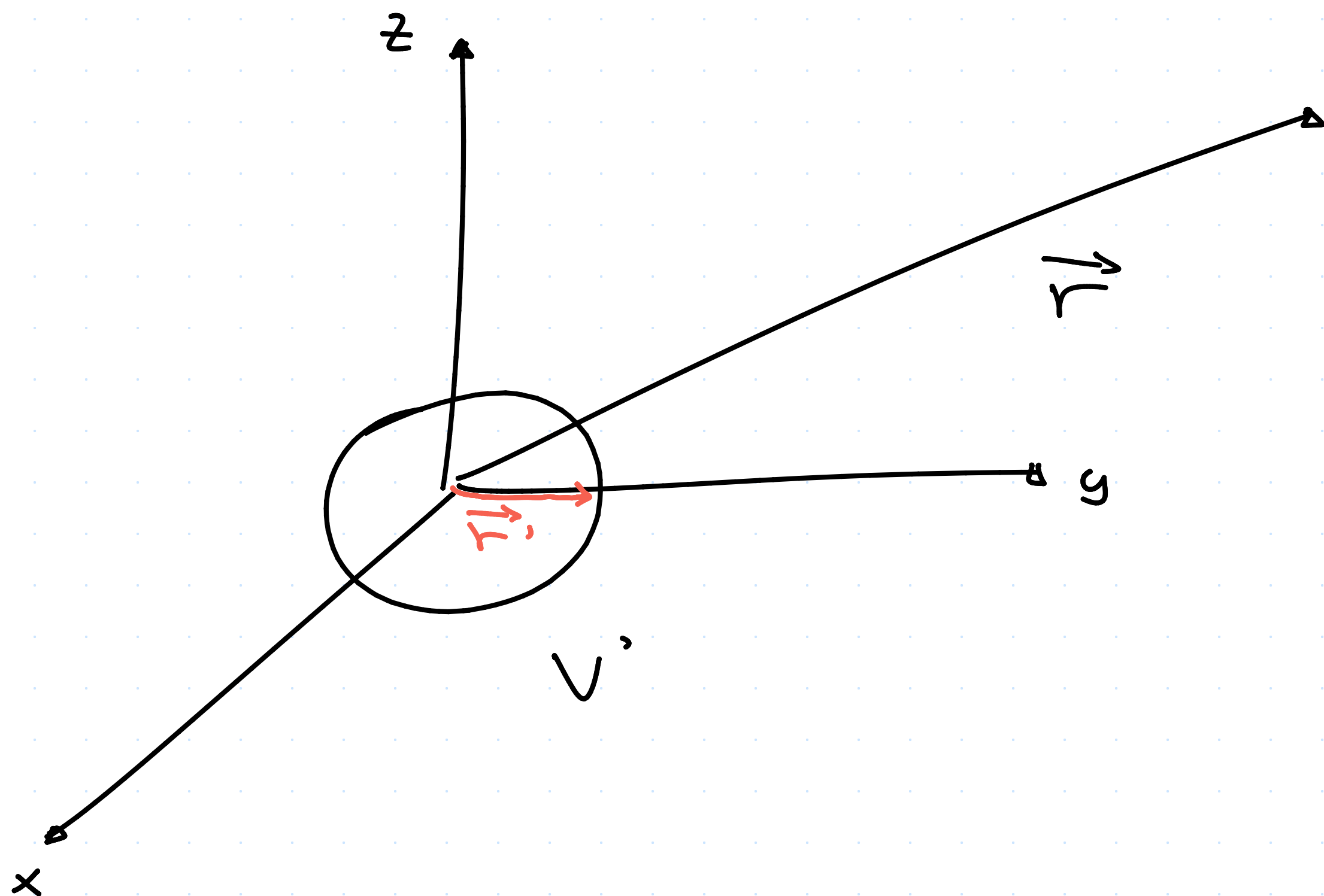
$$\hookrightarrow \int_{V'} \mathbf{r}' \rho(\mathbf{r}') dV'$$

EC. 2-48

Comenzamos con la ecuación 2-45

$$U(r) = \frac{1}{4\pi\epsilon_0} \int_{V'} \frac{\rho(\mathbf{r}')}{|\vec{r} - \vec{r}'|} dV'$$

Tenemos la siguiente distribución de cargas:



$$|\vec{r}'| \ll |\vec{r}|$$

Desarrollando tenemos... calculando el producto punto

$$|\vec{r} - \vec{r}'|^2 = (\vec{r} - \vec{r}') \cdot (\vec{r} - \vec{r}') = \left(|\vec{r}|^2 - 2\vec{r}' \cdot \vec{r} + \vec{r}' \cdot \vec{r}' \right)^{-1/2}$$

$$= |\vec{r}|^{-1} \left\{ 1 - \frac{2 \vec{r} \cdot \vec{r}'}{|\vec{r}|^2} + \frac{r'^2}{r^2} \right\}$$

Resultado correcto

$$|\vec{r} - \vec{r}'|^{-1} = (r^2 - 2 \vec{r} \cdot \vec{r}' + r'^2)^{-1/2}$$

$$|\vec{r} - \vec{r}'|^2 = r^2 - 2 \vec{r} \cdot \vec{r}' + r'^2$$

$$\Rightarrow |\vec{r} - \vec{r}'|^{-1} = \frac{1}{r} \left\{ 1 - \frac{2 \vec{r} \cdot \vec{r}'}{r^2} + \frac{r'^2}{r^2} \right\}^{-1/2}$$

Tenemos

$$|\vec{r} - \vec{r}'|^{-1} = \frac{1}{r} \left\{ 1 - \frac{1}{2} \left(-2 \frac{\vec{r} \cdot \vec{r}'}{r^2} + \frac{r'^2}{r^2} \right) + \left(-\frac{1}{2} \right) \left(-\frac{3}{2} \right) \left(\frac{1}{2!} \right) \left(-2 \frac{\vec{r} \cdot \vec{r}'}{r^2} + \frac{r'^2}{r^2} \right)^2 + \dots \right\}$$

Calculamos

$$\frac{r'}{r} = \frac{|\vec{r}'|}{|\vec{r}|}$$

Expandiendo

$$\begin{aligned} & \left(-2 \frac{\vec{r} \cdot \vec{r}'}{r^2} + \frac{r'^2}{r^2} \right)^2 \\ &= \frac{4 (\vec{r} \cdot \vec{r}')^2}{r^4} - 4 \frac{\vec{r} \cdot \vec{r}'}{r^2} \frac{r'^2}{r^2} + \frac{r'^4}{r^4} \end{aligned}$$

Tenemos

$$|\vec{r} - \vec{r}'| = \frac{1}{r} \left\{ 1 - \frac{1}{2} \left(-\frac{2\vec{r} \cdot \vec{r}'}{r^2} + \frac{r'^2}{r^2} \right) + \left(-\frac{1}{2} \right) \left(-\frac{3}{2} \right) \left(\frac{1}{2!} \right) \right.$$

$$\left. \left(-\frac{2\vec{r} \cdot \vec{r}'}{r^2} + \frac{r'^2}{r^2} \right)^2 + \dots \right\}$$

• verificar ecuación
(2-47)

$$4 \frac{(\vec{r} \cdot \vec{r}')^2}{r^4} \propto \frac{(r r')^2}{r^4} = \frac{r'^2}{r^2}$$

multipole expansion

$$U(\vec{r}) = \frac{1}{4\pi\epsilon_0} \int_{V'} \left[\frac{1}{r} + \frac{\vec{r} \cdot \vec{r}'}{r^3} + \frac{1}{2} \left[\frac{3(\vec{r} \cdot \vec{r}')^2}{r^5} - \frac{r'^2}{r^3} \right] \rho(\vec{r}') dV' \right]$$

Tenemos

$$\vec{r} \cdot \vec{r}' = \sum_{i=1}^3 x_i x'_i$$

$$\rightarrow (\vec{r} \cdot \vec{r}')^2 = \left(\sum_{i=1}^3 x_i x'_i \right) \left(\sum_{j=1}^3 x_j x'_j \right) = \sum_{i=1}^3 \sum_{j=1}^3 x_i x'_i x_j x'_j$$

Último término:

$$\frac{r'^2}{r^3} \frac{r^2}{r^2} = \frac{r'^2}{r^5} \left[r^2 \right] ; \quad r^2 = \sum_{i=1}^3 x_i x_i$$

$$= \sum_{i=1}^3 \sum_{j=1}^3 x_i x_j \delta_{ij}$$

Analiza este procedimiento.